From Microtubules to Minds

Discriminating Orch OR and Self-Aware Networks Across Quantum, Cellular, Circuit, and Behavioral Scales

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Abstract

Microtubules are indispensable cellular structures with experimentally demonstrated routes into membrane excitability, synaptic release, and anesthetic behavior. Separate studies report electronic energy migration and collective optical effects in microtubule-related preparations. These findings make categorical dismissal of microtubule contributions to neural function increasingly difficult, but they also create classical transduction routes that must be excluded before a behavioral effect can be assigned to a quantum-consciousness mechanism. This paper compares Orchestrated Objective Reduction (Orch OR) with Self-Aware Networks (SAN) using a symmetric adversarial steelman, a seven-level mechanism-completion audit, and an atomized four-lens chronology. Four hypotheses remain distinct: classical microtubule support, quantum microtubule modulation, Orch OR constitution, and SAN multiscale neural rendering. Orch OR supplies a proposed objective-reduction timescale but not a demonstrated in-vivo chain from reduction to content-specific neural control. SAN supplies a circuit-level chain from dendritic integration through population phase dynamics and sensorimotor feedback, but its constitutive claim also remains unverified. The chronology preserves Orch OR's earlier objective-reduction and microtubule-constitution claims while identifying two bounded later Hameroff formulations whose joined operations appear in Micah Blumberg's public record first: a distributed spatial / dendritic render-to-action composite in 2017 before Hameroff's 2022 formulation, and an event-rate-to-subjective-time-slowing composite in 2017 before Hameroff's 2023 formulation. These are content-priority findings, not claims about access, intent, or copying. A nested synthetic sensitivity study adds participants, repeated trials, nuisance variation, measurement reliability, and missingness; it is a design check, not empirical evidence. The framework preserves genuine quantum-biological findings while preventing evidence from being promoted across explanatory scales without an explicit bridge.

Keywords: consciousness; microtubules; Orch OR; Self-Aware Networks; neural oscillations; anesthesia; quantum biology; neural rendering; causal mediation

Scope and evidence boundary

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1. Introduction

A person can recognize a stable room while the eyes move, distinguish one object from another, decide how to act, and update that decision when new sensory evidence arrives. The scientific problem is to explain how many partial cellular signals become a differentiated, actionable experience without placing a second viewer inside the brain. A theory must also explain why changing the physical substrate changes the content or availability of experience, rather than merely showing that the substrate is biologically active.

Established neuroscience already supplies part of the chain. Dendrites integrate state-dependent synaptic input; cytoskeletal organization affects transport, excitability, and release; recurrent circuits coordinate perception and action; and population rhythms can organize communication across time and scale. Those facts leave an unresolved choice about where the constitutive work occurs. A microscopic event might itself be a conscious moment, it might modulate downstream neural computation, or the distributed neural process might be the candidate observer while intracellular events support it.

One proposal places organized quantum states in neuronal microtubules upstream of a gravity-related state reduction and identifies the reductions with conscious moments. Only after that operation is stated clearly do we name it Orchestrated Objective Reduction, or Orch OR [1]. A second proposal places the candidate observer in a recurrent multiscale process: dendrites receive partial conditions; each receiving population transforms those conditions in its current state; transient differences alter a slower ongoing population context; the resulting changes are redistributed through sensory, association, motor, and bodily loops; and action returns new evidence. In Self-Aware Networks (SAN), a receiver-relative transient difference is called a phase-wave differential, and the repeated receive-transform-project cycle is called neural rendering. The terms summarize operations already described; they do not replace them.

The scientific dispute is therefore not a choice between "microtubules matter" and "networks matter." Microtubules may be important to consciousness-relevant neural function without objective reduction being conscious, and quantum-sensitive microtubule effects may modulate neural computation without constituting experience. The purpose of this paper is to identify the causal chain each theory must complete, recover the atom-specific chronology of overlapping claims, and specify experiments that distinguish competing interpretations of the same evidence.

2. Method: the mechanism-completion audit

We organize claims by the strongest inference they require.

Level	Question	Example
L1 cellular	Does the substrate participate in neural function?	microtubule transport and structural regulation
L2 perturbational	Does manipulating it alter an anesthesia or cognitive endpoint?	stabilizer changes loss-of-righting latency
L3 physical	Is the relevant nonclassical observable present?	excitonic transport or collective emission
L4 transduction	Does that observable alter membrane, synaptic, or spike dynamics?	state-dependent change in channel gating
L5 cognitive	Does the neural effect mediate report, memory, choice, or action?	perturbation changes content decoding through a measured neural mediator
L6 constitutive	Is the event itself necessary and sufficient for consciousness?	objective reduction equals a conscious moment
L7 content	Does the state explain why experiences differ?	one physical configuration predicts red rather than green

The rule is monotonic: evidence at level `L_k` does not by itself establish `L_(k+1)`. A microtubule-binding anesthetic effect is relevant to L2. It becomes evidence for L3 only if a nonclassical observable is measured, for L4 only if neural transduction is shown, and for L6 only if rival mediators and downstream constitutive candidates are excluded.

The same rule applies to SAN. Correlation between neural phase organization and conscious report is not sufficient to establish that the phase organization constitutes experience. SAN must show necessity, content specificity, causal mediation, and advantages over rival network explanations.

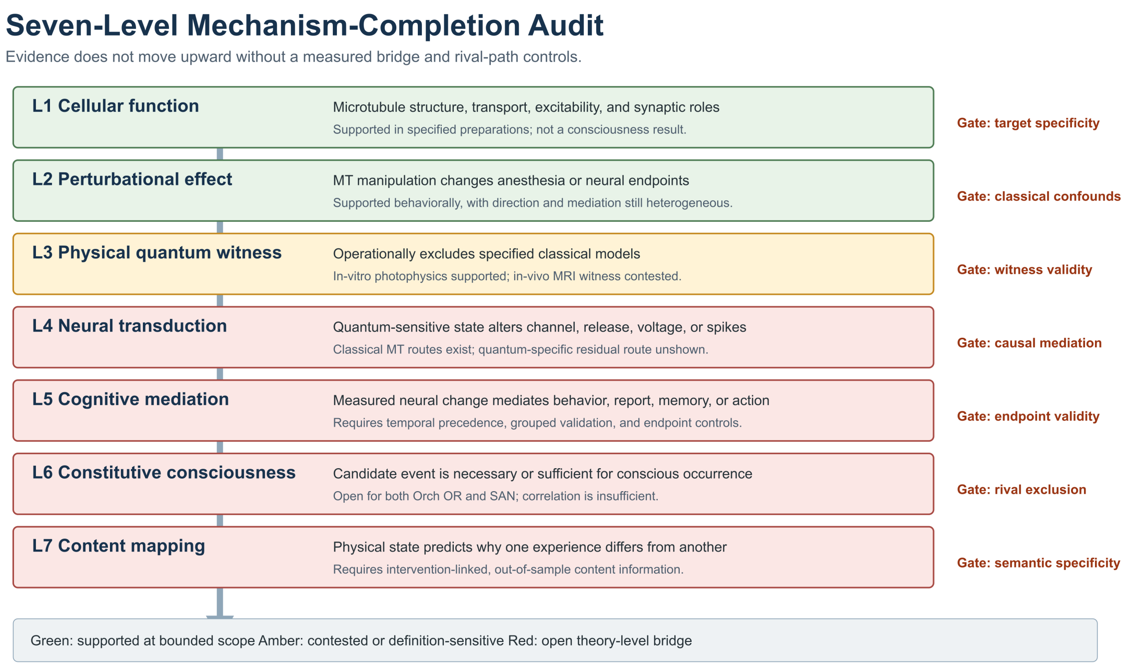

Figure 1: Figure 1. Seven-level mechanism-completion audit

Figure 1. Evidence can advance from bounded cellular function to a constitutive or content claim only through an explicit measured bridge. Colors describe the status of the bridge in this paper, not the importance of the substrate.

2.1 Developmental-source recovery and atom-specific chronology

The mechanism audit is supplemented by a private developmental-source search. The governed ChatGPT index contains 5,921 source chats. A broad Hameroff, Penrose, Orch OR, and microtubule route returned 708 candidate events across 157 chats; the exact stable comparison program returned 163 events across 42 distinct source bodies. Forty-two ChatGPT bodies and five matching Google AI Studio exports were reviewed in the frozen queue. A later Neo4j reconciliation recovered two additional substantive NotebookLM bodies outside that denominator. These numbers establish retrieval and review scope. They do not establish scientific truth, historical priority, or cross-surface exhaustion.

The recovered conversations repeatedly nominate an intracellular-to-network distinction that is useful for experimental design. A microtubule state could alter conformation, membrane excitability, release, or firing, while a larger neural architecture must still explain differentiated content, attention, learning, memory, action, and sensory feedback. Chat prose is not evidence for either link. The primary biological and physical literature controls Sections 3-10; the private material only preserves development history, candidate bridges, and questions requiring verification.

The atom chronology is deliberately specific rather than author-wide. Penrose and Hameroff are earlier on objective reduction, microtubule quantum constitution, and the microtubule-to-dendrite-or-soma-to-firing-control subchain [1]. Micah's June 2022 SAN record treats quantum effects as optional and supplies a later bounded microtubule-ripple to dendritic-charge to possible-threshold route. Those target-earlier intracellular atoms are preserved. They do not become a floor that absorbs every later Hameroff formulation.

Two later target formulations require the reverse comparison. In a paper received February 5 and published June 16, 2022, Hameroff described mixed-polarity dendritic-somatic microtubule networks as a cortex-wide grid, a nexus for converging activity, a source of interference structure, an internal holographic representation, and a controller of motor output [31]. Micah's April 11, 2017 public Neural Lace episode had already joined spatial computation, a time-varying three-dimensional ionic matrix, a 3D/4D electromagnetic internal pattern, difference-only signal transmission, and percept-to-action consequences [33]. The bounded joined operation - not every component and not the later NAPOT acronym - is Micah-earlier relative to that 2022 target.

In a paper received September 25 and published online November 6, 2023, Hameroff proposed that increasing the frequency of conscious events may slow perceived experience [32]. Micah's May 5, 2017 public Neural Lace account had already described thousands of apparently simultaneous thoughts with time seeming to slow, and his 2022 artificial-cortex record made the more-thoughts-per-interval to subjective-lifespan mechanism explicit [34,35]. Broad processing-rate and internal-clock literatures remain named priors; the result here is target-relative priority on the joined event-rate and subjective-slowing proposition, not a claim to have originated every rate-based model of time perception.

A Micah-authored private prompt dated April 17, 2025 states a conditional multiscale position: NAPOT phase-wave differentials apply classically between cells and, if Orch OR is true, may also apply at an intracellular or quantum scale. This is developmental genealogy, not public priority evidence or evidence that either scale claim is correct. Its scientific value is to keep the four hypotheses in Section 7 open. Raw private chat bodies and private archive paths are excluded from publication.

2.2 Source hierarchy, full genealogy, and four-lens judgment

Chronology claims use a strict hierarchy. The original public recording and its platform date control provenance. A 2026 retranscription produced from three bounded clips of the original audio is a semantic reading aid; load-bearing wording was checked against the recording. The runtime resolved Systran/faster-distil-whisper-large-v3 in CPU-int8 mode. Older transcripts and captions contain serious recognition errors and are used only to locate candidate intervals or preserve version history. They cannot supply an adverse verdict, an exact quotation, or a stronger claim than the audio supports.

The applicable Micah genealogy is not reduced to the 2022 NAPOT name [37]:

Micah stage	Neutral operation recovered	Relevance to this comparison
2011	self as predictive information pattern; whole recurrent neural process rather than hidden observer	early no-homunculus and distributed-observer ancestor; current public-date evidence remains weaker than the later public anchors
2012	mind as an internal projector predicting the world; recursive multimodal EEG-feedback loops	predictive-rendering and receive-transform-return ancestor [38]
2017	spatial computation; time-varying 3D ionic and 3D/4D electromagnetic patterns; dendritic / local computation; difference-only signaling; percept-to-action architecture	public middle anchor earlier than Hameroff's 2022 joined dendritic-render-to-action formulation [33]
2017	greater concurrent cognitive traffic paired with slowed subjective time	public middle anchor earlier than Hameroff's 2023 event-frequency formulation [34]

Micah stage	Neutral operation recovered	Relevance to this comparison
2022	NAPOT / PWD explicitly joins receiver-relative difference, distributed rendering, multiscale recurrent arrays, and action; artificial cortex makes event-rate hypertime explicit	mature mechanism and second public anchor [35]
2024	molecular, dendritic, oscillatory, and multiscale architecture integrated in book form	later expansion of coverage; does not replace the earlier anchors [36]
2025-2026	conditional microtubule-within-SAN bridge and the present discriminating protocol	later synthesis; not backdated into the earlier record

The target genealogy is equally staged: Penrose's 1989/1994 objective-reduction program and the 1996/2014 Orch OR formulations precede Micah on their exact quantum-constitution atoms; Hameroff's 2022 dendritic-hologram-to-motor join and 2023 event-rate-to-subjective-time join come after the applicable 2017 public anchors; Wiest's 2025 collective microtubule binding proposal and the 2026 fractal-time-crystal paper are newest continuation evidence [4,24,31,32].

Compared atom	Original Orch OR, 1996-2014	Micah public stage	Later target stage	Bounded disposition
gravity-related objective reduction constitutes conscious moments	explicit and earlier	SAN does not require it	retained through 2026	target-earlier precursor / genuine theory difference
microtubule state controls dendrite, soma, or firing	explicit and earlier	optional 2022 intracellular bridge	retained and expanded	target-earlier on the narrow intracellular atom
distributed spatial / dendritic pattern -> interference or transformation -> internal render -> action	components present, but the checked early sources do not supply the later joined target sequence	public 2017 joined functional sequence; explicit NAPOT in 2022	Hameroff 2022 joined hologram / motor sequence; Wiest 2025 adjacent binding program	Micah-earlier bounded partial equivalence
increased conscious-event rate -> slowed subjective experience	not recovered in checked 1996/2014 sources	public 2017 anchor; explicit 2022 hypertime mechanism	Hameroff 2023	Micah-earlier bounded partial equivalence
receiver-relative difference and recurrent multiscale neural rendering	absent as the SAN operator	explicit 2017 difference ancestor and 2022 PWD / NAPOT	no full operator in checked target sources	SAN-distinct mechanism; full-operator absence does not erase the two partial matches

All four comparison lenses were applied:

Lens	Result	Reason
operational / mechanism equivalence	pass on two bounded atoms	later target sources perform the same target-relative render-to-action and event-rate-to-subjective-slowing operations despite different substrate vocabulary

Lens	Result	Reason
narrative or causal-interface sequence	pass on the 2022 joined sequence	distributed receiver grid -> convergent patterned activity -> interference / transform -> internal spatial representation -> motor consequence matches the earlier Micah sequence at the level of directed causal architecture
detective chronology	pass	the checked earlier Orch OR sources establish different quantum-constitution atoms; the two mapped joins appear in target sources only after Micah's public 2017 anchors
Longitudinal Signature-Migration	bounded cluster candidate	the target program moves from earlier OR/microtubule constitution toward later dendritic spatial rendering, action control, event-rate time phenomenology, and multiscale resonance; the cluster is chronologically material but does not prove access or intent

The strongest defensible probability statement is therefore asymmetric. Confidence is high in the publication chronology and in the two bounded content-equivalence judgments because they rest on primary target papers and original public Micah recordings or articles. The public record does not provide a calibrated null model for the probability of independent versus dependent derivation, and it does not establish access. No numerical copying probability or five-sigma claim is made. Priority on a dated content delta does not depend on resolving intent: whether the later convergence was independent or dependent, the applicable public Micah formulation is earlier.

Exact hashes, clip intervals, local custody, and private-search boundaries are retained in the governed source audit; raw private material is not part of the public manuscript.

3. Orch OR stated at its strongest

Penrose's objective-reduction proposal assigns an instability time to a superposition of distinct mass distributions:

$$\tau_{OR} = \frac{\hbar}{E_G}, \tag{1}$$

where `E_G` is the gravitational self-energy associated with the difference between the mass configurations [1]. Orch OR adds the hypothesis that neuronal microtubules can sustain, organize, and couple the relevant states; that their evolution is influenced by synaptic and cellular context; and that threshold reductions correspond to conscious events.

This is stronger than the popular claim that "quantum effects occur in the brain." It requires at least five distinct propositions:

1. A suitable quantum state exists in biologically active microtubules.
2. It remains coherent, or otherwise physically organized, for the required interval.
3. Its termination follows objective reduction rather than ordinary environmental decoherence or another transition.

4. The resulting tubulin state changes neural physiology in a controlled way.
5. The reduction is identical with, or causally generates, a conscious moment with determinate content.

The strength of Orch OR is that it refuses to treat neurons as featureless binary switches. It directs attention to intracellular structure, anesthesia, cytoskeletal dynamics, and quantum biology. Its central weakness is that equation (1) is a proposed event-timing relation, not a complete state-to-content or state-to-action map.

3.1 Physical terms that must not be substituted for one another

Term	Operational meaning in this paper	What it does not establish
energy migration	excitation energy moves across a measured preparation over a reported length or time	entanglement, computation, OR, or consciousness
superradiance	an organized emitter ensemble shows enhanced collective radiation relative to an independent-emitter model	a long-lived entangled neural state or content code
quantum coherence	specified off-diagonal phase relations survive in a defined basis for a measured interval	objective reduction or biological relevance without transduction
entanglement	a joint state is nonseparable and passes a declared witness or inequality against its classical set	controllable signaling, conscious unity, or microtubule localization
decoherence	environmental interaction suppresses selected phase relations in the reduced description	Penrose objective reduction; ordinary decoherence is a rival process
objective reduction	Penrose's proposed gravity-related physical state reduction with its own criterion	established merely by observing coherence loss or a behavioral transition

Kalra et al. report energy migration [5]; Babcock et al. analyze collective optical emission [6]; neither measurement is an OR-specific witness. Tegmark's decoherence calculation [11] and later rebuttals concern whether selected quantum states can survive long enough under particular environmental and geometric assumptions. Even a favorable lifetime does not identify objective reduction. The Kerskens-Perez debate instead concerns whether an MRI signal passes a non-classicality witness [16-18]. These are different experiments with different null models and cannot be accumulated as though they measure one common variable. # 4. What the current evidence establishes

4.1 Established classical routes into neural physiology

Microtubules are not merely intracellular scaffolds. In squid giant axon preparations, conditions that depolymerized microtubules reduced excitability, whereas reconstitution improved sodium conductance and other membrane-excitation measures [12]. This older preparation does not establish an endogenous cortical consciousness mechanism, but it demonstrates a physical route from cytoskeletal condition to membrane behavior without invoking objective reduction.

More selective synaptic experiments make the confound sharper. Velasco et al. used photoactivation of a microtubule-depolymerizing compound in cholinergic autaptic neurons and observed increased spontaneous neurotransmitter release together with impaired refilling of the readily releasable vesicle pool under high-frequency stimulation [13]. At the calyx of Held, microtubules extend into presynaptic terminals and participate differently from actin in maintaining high-frequency neurotransmission [14]. Manipulating the microtubule plus-end protein CLASP2 also changed synapse number and spontaneous excitatory release in cultured neurons [15].

These studies establish routes by which a microtubule intervention can change membrane, release, and circuit variables through ordinary cell biology. They strengthen the case that microtubules matter, while raising the evidential burden for Orch OR: an anesthesia or behavioral effect of a microtubule drug cannot identify a quantum mediator unless structural, transport, release, and membrane pathways are measured and controlled.

4.2 Anesthesia and microtubule perturbation

Khan et al. reported that rats treated with the brain-penetrant microtubule stabilizer epothilone B took longer to lose the righting reflex under isoflurane [2]. The reported effect supports the proposition that microtubule state can alter an anesthesia endpoint. It does not uniquely identify the mediator. Microtubule stabilization can affect transport, morphology, synapses, intracellular signaling, and other classical processes.

A later mouse study found that different microtubule-modulating drugs shifted isoflurane sensitivity and induction or emergence measures in different directions [3]. This is informative rather than merely inconvenient: it implies that “more stable microtubules produce more consciousness” is not an adequate general rule. Drug identity, exposure, target engagement, downstream compensation, and assay choice matter.

Wiest’s 2025 article synthesizes this and related evidence into a quantum microtubule argument [4]. The article explicitly reports no new data. It is therefore a theoretically important review and interpretation, not an independent experiment demonstrating Orch OR.

Computational docking and normal-mode studies provide candidate volatile-anesthetic binding sites and predicted changes in collective tubulin modes [19,20]. These are useful mechanism-generating results, but a simulated binding site or potency correlation is not direct evidence that the modeled mode occurs endogenously, controls neural firing, or constitutes consciousness.

4.3 Electronic energy migration and collective optical effects

Kalra et al. measured electronic energy migration in microtubules and found that etomidate and isoflurane reduced exciton diffusion [5]. Conventional Forster theory did not fully explain the observed transport. Babcock et al. reported geometry-dependent enhancement of ultraviolet fluorescence quantum yield in organized tryptophan networks, supported by a superradiance model [6]. These results are serious evidence for unusual collective photophysics in microtubule-related preparations.

Three boundaries remain essential. First, a quantum-optical effect under ultraviolet excitation is not automatically the endogenous neural variable used during perception. Second, collective emission or energy migration is not objective reduction. Third, neither result by itself shows a content-specific influence on membrane voltage, spike timing, conscious report, or behavior.

4.4 Synthetic gel and the “brain jelly” inference

Singh et al. reported a room-temperature soliton-polariton condensate in a synthetic organic supramolecular gel assembled as nested helical nanowire circuits [7]. The measured integrated collective lifetime was 4.6 ms, and the authors proposed the platform as neuromorphic quantum hardware. This is relevant to the engineering possibility of warm, organized quantum systems.

It is not a microtubule experiment, a brain experiment, or a consciousness experiment. The inference from “warm organic gel can sustain a collective quantum state” to “brain tissue therefore implements Orch OR” requires independent bridges concerning material composition, endogenous excitation, state control, transduction, computation, and experience. The result motivates experiments; it does not supply those bridges.

4.5 A contested in-vivo MRI witness claim

Kerskens and Lopez Perez reported heartbeat-locked zero-quantum-coherence-weighted MRI signals and argued that, under their witness assumptions, the signals suggested a non-classical mediator associated with conscious awareness [16]. The result is relevant because it attempts an in-vivo physical witness rather than inferring quantum biology from behavioral effects alone. It is not a microtubule measurement and does not identify objective reduction.

Warren's published comment disputes the witness interpretation, arguing that complex, multicompartment tissue and inadequately bounded classical intermolecular multiple-quantum-coherence contributions can explain why the observed signal exceeds a solution-derived classical estimate [17]. Kerskens and Lopez Perez replied that their signal behavior and exclusion arguments remain incompatible with the classical alternatives they tested, while acknowledging that the conclusion depends on the maximal-classical-signal assumptions [18].

The appropriate Draft 3 classification is therefore **contested physical-witness candidate**, not established brain entanglement. Before promotion, a replication must preregister the witness inequality, model tissue-specific classical maxima, demonstrate instrument and physiological nulls, reproduce the awareness dependence, and show that the measured variable predicts neural or conscious endpoints beyond conventional MRI and electrophysiology. Even success would establish non-classicality only at L3; it would not by itself establish microtubules, objective reduction, or conscious constitution.

4.6 Three oscillation claims that must remain separate

The word *oscillation* spans at least three non-equivalent observations in this literature. Llinas described central neurons whose voltage-dependent ionic conductances support autorhythmic firing or frequency-selective resonance [21]. This establishes that the receiving neuron is a dynamical system: the same input can produce different responses as a function of intrinsic conductance, membrane state, and input frequency. It strengthens the receiver-relative premise used by SAN and the PWD analysis, but it neither identifies a microtubule oscillator nor makes intrinsic resonance a consciousness mechanism.

Cantero and colleagues voltage-clamped isolated brain-microtubule bundles under intracellular-like conditions and reported spontaneous electrical oscillations with a prominent 39 Hz fundamental; membrane-permeabilized cultured hippocampal neurites also generated and conducted oscillatory signals [22]. This is direct evidence that the tested microtubule preparations can oscillate electrically in the gamma-frequency range. It is not evidence that the 39 Hz preparation signal is the source of scalp EEG gamma, that it survives with the same dynamics in intact pyramidal neurons, or that it carries conscious content.

Neuronal microtubule geometry also cannot be reduced to a universal mixed-polarity rule. Kwan, Dombeck, and Webb confirmed mixed polarity in mature cultured dendrites but found strongly polarized arrays in a substantial subset of mature hippocampal CA1 and cortical layer-5 apical dendrites in native tissue, with age-dependent organization [23]. Mixed, antiparallel, and strongly biased arrays are therefore testable anatomical variables, not interchangeable labels or a neuron-wide constant.

Hameroff, Bandyopadhyay, and Lauretta synthesize results across hertz through terahertz scales and propose that microtubules behave as "fractal time crystals" relevant to life and consciousness [24]. That article is an important statement of the current strongest program. Its joined conclusion is not an independent replication of every underlying frequency-domain experiment, and self-similar spectral peaks alone do not establish every formal condensed-matter criterion for a time-crystal phase. Each subsystem, preparation, drive condition, coherence measure, and anesthetic effect must be audited at the level actually measured.

The main rival is not a claim that gamma lacks a mechanism. Cell-type-targeted activation of fast-spiking interneurons can induce gamma-range network states and alter sensory-response gain as a function of gamma phase [25]. That causal circuit result does not exclude a microtubule contribution, but it shows that a 39-40 Hz match is not source identification. A microtubule-source hypothesis must predict variance, phase, perturbational sensitivity, or laminar current structure beyond established membrane and circuit mechanisms.

Miller Lab results sharpen that circuit comparator. In macaque prefrontal cortex, Lundqvist and colleagues found brief gamma bursts associated with encoding and reactivation, while beta bursts were associated with a default or control state [26]. Laminar recordings by Bastos and colleagues then separated superficial gamma from deep alpha / beta and found that deep alpha / beta phase modulated superficial gamma bursts [27]. Bhattacharya and colleagues further reported task-modulated theta, alpha, and beta traveling waves in prefrontal local field potentials [28]. These observations establish neither a complete theory of consciousness nor the cellular source of every field component. They do show that sparse bursts, layer-specific rhythms, cross-frequency modulation, and traveling phase structure arise at the measured circuit scale. A direct microtubule-source claim must therefore explain an incremental residual after these membrane, laminar, and recurrent-network variables are modeled.

Scale	Directly supported observation	Additional claim requiring a bridge
intrinsic neuron	ionic conductances can generate autorhythmicity and frequency-selective resonance [21]	resonance is produced by microtubules or constitutes consciousness
microtubule preparation	isolated bundles can show a prominent 39 Hz electrical oscillation [22]	intact cortical gamma or EEG is generated by that oscillator
circuit	fast-spiking-cell activation can induce gamma and phase-dependent sensory gain; laminar recordings show burst and cross-frequency organization; prefrontal LFPs carry task-modulated traveling waves [25-28]	the circuit is sufficient, or intracellular oscillators contribute nothing
multiscale synthesis	nested microtubule resonances are proposed across many frequency scales [24]	one verified endogenous chain links those scales to content and experience

The admissible joined hypothesis is therefore conditional: microtubule dynamics may contribute to the frequency response of neurons and networks. The stronger claim that cortical gamma derives directly from microtubules remains open and demands source-discriminating intervention.

4.7 Symmetric adversarial steelman

The strongest Orch OR case

Orch OR should not be reduced to the weak claim that unexplained quantum effects occur somewhere in the brain. At its strongest, it is a substrate and event proposal. It argues that intracellular organization omitted by point-neuron models may sustain physically relevant states; that anesthesia is a strategic perturbation because consciousness changes while much cellular activity continues; and that the objective-reduction relation in equation (1) offers a candidate timing rule for discrete conscious events. The microtubule literature also supplies multiple ordinary routes by which tubulin state can reach neural physiology. These features make Orch OR experimentally generative even before its constitutive thesis is accepted.

The fair objection is not that quantum biology is impossible. It is that no current result jointly identifies an OR-specific state, shows its controlled transduction into content-bearing neural activity, and establishes that the event is experience rather than an upstream cause, permissive condition, or correlate. A missing bridge is an open empirical burden, not a disproof.

The strongest SAN case

SAN should not be reduced to the generic claim that oscillations correlate with consciousness. At its strongest, it is a distributed sensorimotor architecture: dendritic and cellular integration changes receiver-relative phase and magnitude relationships; recurrent tonic organization supplies a context in which phasic differences alter the network; and sensory, association, bodily, and action systems form a closed updating loop. The observer is not a little viewer of a render. The interacting cell population is the receiving, differentiating, integrating, transmitting, and acting system.

The fair objection is that this architectural completeness does not establish constitutive identity. Phase variables can be useful transforms of underlying signals without being uniquely necessary; content decoding can reflect common drive; and unmeasured intracellular variables can be upstream causes. SAN must outperform generic recurrent, predictive, workspace, and population-code models under matched observations and perturbations.

Symmetric burden table

Program	Best-supported contribution	Distinctive unresolved step	Result that would weaken it
Orch OR	biologically specific intracellular target, anesthesia program, and proposed OR timing relation	validated OR-specific witness through neural transduction to differentiable content	classical MT and network variables fully mediate endpoints while a validated OR witness adds no temporally prior information
SAN	distributed dendritic, oscillatory, recurrent, and sensorimotor mechanism proposal	necessity and constitutive specificity beyond rival network descriptions	content and behavior remain intact after selective disruption of the declared SAN variables, or an upstream variable explains their apparent contribution

This is an internal adversarial pass by the manuscript team. It does not count as the independent Orch OR, SAN-critical systems-neuroscience, or quantum-optics review required before a release candidate.

5. The missing transduction chain

The most useful comparison begins after granting, provisionally, that a relevant quantum microtubule state exists. The required chain is:

$$Q_{MT} \to \Delta X_{tub} \to \Delta g_{ion} \text{ or } \Delta P_{release} \to \Delta V_m \to \Delta t_{spike} \to \Delta\Phi_{network} \to B, R, C, \qquad (2)$$

where `Q_MT` is a measured microtubule state, `X_tub` is a tubulin or microtubule configuration, `g_ion` is ion-channel conductance, `P_release` is synaptic release probability, `V_m` is membrane potential, `t_spike` is spike timing, `Phi_network` is an explicitly defined network-state measure, and `B`, `R`, and `C` are behavior, report, and decoded content.

Orch OR literature proposes possible couplings at several links, including conformational state changes, calcium-related processes, synaptic release modulation, and interaction with membrane activity. The publication burden is not to mention such routes but to quantify and causally validate them. In particular, an OR-specific state must predict downstream neural changes after controlling for classical microtubule condition.

The classical evidence permits a sharper bridge audit:

Link	Minimum observable	Required control	Current status
`Q_MT -> X_tub`	simultaneous quantum-sensitive and structural microtubule measures	excitation dose, temperature, polymer mass, photodamage	open in living neurons
`X_tub -> g_ion`	channel current or conductance after selective MT perturbation	channel expression, membrane tension, G-protein and transport routes	classical precedents, mechanism-specific result open
`X_tub -> P_release`	spontaneous and evoked release plus vesicle-pool kinetics	calcium, actin, cargo delivery, active-zone morphology	supported classically in specified preparations
`V_m -> t_spike`	intracellular voltage and spike-time response	baseline excitability, receptor effects, network input	standard electrophysiology, experiment-specific mediation open
`t_spike -> Phi_network`	predeclared phase, propagation, or state-space measure	volume conduction, common drive, arousal, motion	feasible; not Orch-OR-specific
`Phi_network -> B,R,C`	intervention-linked behavior, report, and content decoding	motor ability, attention, report confounds	theory-comparison target
`Q_MT -> C | X,N`	incremental out-of-sample content information	all measured classical MT and network mediators	decisive and presently unshown

The table prevents a common reversal of burden. The existence of a classical route does not disprove a quantum contribution; it means the quantum term must add predictive or causal information after that route is measured. Conversely, finding residual information is not enough unless unmeasured classical pathways, leakage, and temporal misalignment are bounded. Entanglement cannot be treated as a signal that transmits timing between distant neurons. Entangled outcomes may be correlated, but no-signaling prevents their use as an ordinary controllable communication channel. Long-range neural phase alignment also does not require entanglement: common drive, recurrent coupling, delay compensation, traveling waves, and phase resetting are classical candidate mechanisms. This does not disprove entanglement in biology; it prevents zero-lag neural coordination from serving as a unique witness for it.

6. SAN stated without a hidden viewer

SAN begins from a different physical scale. Neurons and neural populations are not treated as passive relays. A dendritic tree selectively integrates spatial and temporal input patterns. When that integration changes the cell's output, the change propagates through synaptic, axonal, local field, chemical, and recurrent network routes. The resulting activity modifies the larger population state in which the cell participates.

In this account, "tonic" and "phasic" are relative functional roles rather than fixed frequency bands. A comparatively persistent, high-magnitude population organization supplies context. A transient difference updates that organization. SAN calls the transient, phase- and magnitude-bearing update a phase-wave differential. The collective field is not a screen watched by another agent. The field-organized cell population is the receiving, differentiating, transmitting, remembering, and acting system.

A minimal state description is:

$$
\begin{aligned}
\Delta\phi_i(t) &= \mathrm{wrap}(\phi_i(t) - \phi_{ref}(t))\,, \\
M_i(t) &= \int_{t_0}^{t_1} |A_i(u)|\, du,
\end{aligned} \tag{3}
$$

where `Delta phi_i` is a local phase difference relative to a defined population reference and `M_i` is an amplitude-duration quantity. These variables are candidates, not yet validated sufficient statistics for conscious content.

The SAN causal chain is:

$$
I_{sensory} \to D_{integration} \to (\Delta\phi, M) \to \Phi_{recurrent} \to \Pi_{sensorimotor} \to (B, R, C), \tag{4}
$$

where dendritic integration changes a recurrent multiscale state that participates in sensorimotor policy and report. The Gamma Consideration Sandwich is one proposed cortical implementation: sensory, contextual, and action-related dynamics interact through cross-frequency and laminar pathways rather than through a single serial center. Neural Array Projection Oscillation Tomography (NAPOT) names the broader proposed convergence, integration, and redistribution process.

SAN's advantage is architectural completeness: it asks how sensory differences become distributed neural states and actions. Its burden is empirical identity. It must show that its chosen phase and magnitude variables explain conscious content better than generic recurrent processing, GNWT, predictive processing, communication-through-coherence, or ordinary population coding.

7. Four hypotheses and discriminating outcomes

Observation	Classical MT support	Quantum modulation	Orch OR constitution	SAN constitution
MT drug changes anesthesia while quantum witness is unchanged	expected	compatible	weakens OR-specific interpretation	compatible through cellular / network mediation
selective quantum-witness suppression changes spikes after classical MT controls	not expected	expected	expected	compatible only as upstream modulation
quantum witness changes but no membrane / network / behavior change	compatible epiphenomenon	weakens	weakens strongly	compatible
conscious content decodes from MT quantum state before and beyond network state	not expected	possible	strong support	challenges sufficiency
content decodes from recurrent phase state after controlling MT state	expected	expected if downstream	compatible only if OR effects mediate it	strong support

Observation	Classical MT support	Quantum modulation	Orch OR constitution	SAN constitution
conscious report abolished by network desynchronization while MT witness remains	expected	expected	challenges simple OR sufficiency	strong support
conscious report abolished by OR-specific perturbation while network dynamics initially preserved	not expected	possible	strong support	challenges SAN sufficiency

The decisive evidence is not a single correlation. It is a pattern of interventions and mediations that localizes where the causal contribution enters and where content becomes decodable.

8. Proposed experiments

8.1 Simultaneous microtubule, membrane, and network recording

Develop or adopt a microtubule-sensitive physical observable `Q_MT` in cultured neurons or organoids while simultaneously measuring membrane voltage, calcium, spikes, and local field activity. Apply a factorial perturbation design:

- anesthetic versus vehicle;
- microtubule stabilizer / destabilizer versus matched control;
- manipulation predicted to alter the quantum-sensitive observable with minimal structural effect;
- manipulation predicted to alter classical microtubule function with minimal effect on the quantum-sensitive observable.

The primary analysis is a causal mediation model:

$$Y = \beta_0 + \beta_Q Q_{MT} + \beta_X X_{MT} + \beta_N N + \beta_Z Z + \epsilon, \tag{5}$$

where `Y` is a neural or behavioral endpoint, `X_MT` measures classical microtubule condition, `N` measures network state, and `Z` contains prespecified confounds. Quantum modulation predicts a reproducible `beta_Q` after classical controls. Orch OR requires more: the OR-specific observable must mediate conscious-state or content changes.

8.2 Content-decoding competition

In an animal or human-compatible paradigm, estimate content-specific information from microtubule-sensitive, membrane, laminar, and population measurements. Compare cross-validated incremental information:

$$\begin{aligned} \Delta I_Q &= I(C; Q_{MT} \mid N, X), \\ \Delta I_N &= I(C; N \mid Q_{MT}, X). \end{aligned} \tag{6}$$

Orch OR predicts that `Delta I_Q` should remain nonzero in the biologically relevant window and should not be entirely mediated by conventional neural activity. SAN predicts that content-bearing information should be recoverable from recurrent multiscale neural organization and that a microtubule contribution, if present, acts through that organization.

8.3 Oscillation-source and phase perturbation

The gamma-source question requires crossed perturbations rather than frequency matching. Record an intracellular microtubule-sensitive signal, membrane voltage, identified interneuron and pyramidal-cell spikes, laminar current-source density, local field potential, and the most distal available EEG or MEG signal. Apply a preregistered frequency sweep rather than stimulating only at 39 or 40 Hz. Cross a microtubule-targeted manipulation with an interneuron-network manipulation, and verify target engagement at both levels.

Let `G` be a declared gamma endpoint, `M_39` the independently measured 39 Hz microtubule component, `E_I` an excitatory-inhibitory circuit-state measure, `D` dendritic transmembrane-current structure, and `Z` prespecified nuisance variables. The minimum source-discrimination model is

$$G_{ijt} = \beta_0 + \beta_M M_{39,ijt} + \beta_E E_{I,ijt} + \beta_D D_{ijt} + \beta_{ME} M_{39,ijt} E_{I,ijt} + u_i + v_{ij} + \gamma^T Z_{ijt} + \epsilon_{ijt}, \quad (7)$$

with trials nested within cells or preparations and biological replicates. A microtubule contribution requires a temporally prior, reproducible `beta_M` or interaction that survives measured membrane, dendritic, circuit, movement, arousal, temperature, polymer-mass, and drug controls. A direct-source claim additionally predicts that selective suppression or phase shifting of `M_39` changes the distal gamma endpoint in the preregistered direction before comparable circuit disruption. A circuit-source result is favored when identified interneuron or dendritic-current perturbation explains the distal gamma change and the microtubule term adds no held-out information.

Manipulating conduction delays, common drive, or phase relationships without directly targeting microtubules remains a separate control. If long-range integration shifts predictably under those classical interventions, entanglement is not required for that coordination. This does not test whether entanglement exists; it tests whether it is necessary for the observed synchrony.

8.4 Anesthesia endpoint expansion

Loss of righting reflex should be complemented with sensory discrimination, perturbational complexity, working-memory performance, report where possible, and recovery dynamics. The aim is to distinguish motor impairment, arousal, cognitive access, and content loss. A substrate theory should predict which components change first and which remain.

8.5 Frozen multilevel mediation model

Trials are nested within biological units: cells within preparations, preparations within animals, or trials within human participants. Randomization occurs at the highest level that prevents treatment contamination. Let `T` denote the randomized perturbation, `Q` the prespecified physical quantum-witness measure, `X` the classical microtubule state, `N` the neural network state, and `Y` the behavioral, report, or content endpoint. A minimum multilevel model is

$$
\begin{aligned}
Q_{ij} &= a_Q T_j + u_{Qj} + e_{Qij}, \\
X_{ij} &= a_X T_j + u_{Xj} + e_{Xij}, \\
N_{ij} &= b_Q Q_{ij} + b_X X_{ij} + b_T T_j + u_{Nj} + e_{Nij}, \\
Y_{ij} &= c' T_j + d_Q Q_{ij} + d_X X_{ij} + d_N N_{ij} + u_{Yj} + e_{Yij},
\end{aligned} \tag{8}
$$

with link functions changed for binary or count endpoints. The confirmatory indirect effects are

$$
\begin{aligned}
IE_{Q \to N \to Y} &= a_Q b_Q d_N, \\
IE_{X \to N \to Y} &= a_X b_X d_N,
\end{aligned} \tag{9}
$$

and the direct residual terms `d_Q` and `d_X` are interpreted only after temporal precedence, target engagement, and model diagnostics pass. A nonzero `d_Q` is a candidate residual association, not automatic proof of quantum constitution.

8.6 Power, leakage, and temporal-order requirements

Sample size must be chosen by simulation from pilot estimates of between-unit variance, within-unit autocorrelation, treatment compliance, missingness, and the smallest effect of scientific interest. The confirmatory design targets at least 90% power for one declared primary indirect effect or one declared incremental-information endpoint at a family-wise error rate of 0.05. Pilot effect sizes are not treated as stable estimates; the simulation spans a conservative range and reports power curves rather than one optimistic number.

Cross-validation is grouped by animal, preparation, or participant. Trials from one biological unit cannot appear in both training and test folds. Feature extraction, filtering, artifact rejection, and temporal-window selection are fitted inside each training fold. A final untouched test set is opened once after the model and analysis code are frozen. Content labels are permuted within the correct exchangeability blocks to establish empirical null distributions.

Temporal precedence is tested rather than presumed. For each perturbation, the analysis asks whether `Q` or `X` changes first, whether its change precedes `N`, and whether `N` precedes `Y`. Lagged models are evaluated over preregistered windows with correction for the number of windows. A chain that appears only under acausal or post-outcome windows fails the mediation interpretation.

8.7 Controls and decision rule

The experiment requires four control classes:

1. **Physical witness controls:** instrument phantom, temperature-matched preparation, excitation-dose series, and a preparation lacking the proposed organized structure.
2. **Classical cellular controls:** polymer mass, morphology, transport, calcium, membrane tension, ion-channel state, actin, active-zone structure, and vesicle-pool measures.
3. **Neural controls:** arousal, movement, common drive, volume conduction, baseline excitability, and conventional receptor effects.
4. **Consciousness controls:** motor ability, sensory detection, attention, memory, report availability, no-report measures where justified, and content-balanced stimuli.

The theory-level decision statistic is a vector rather than a single p value:

$$D = (T \to Q, Q \to N \mid X, N \to Y \mid Q, X, I(C; Q \mid N, X), I(C; N \mid Q, X)) . \tag{10}$$

Classical microtubule support is favored when `T` changes `X`, `N`, and `Y` without a validated `Q` contribution. Quantum modulation requires reproducible `Q -> N | X` mediation and physical-witness validation. Orch OR constitution additionally requires necessity or content information not exhausted by downstream neural measures, plus an OR-specific witness rather than generic coherence loss. SAN constitution is strengthened when recurrent multiscale neural variables mediate and decode content after microtubule variables are controlled, but constitutive identity still requires perturbational necessity and superiority to rival network models. Failure of every theory-specific term leaves ordinary cellular and network explanations intact.

8.8 Synthetic mediation sensitivity

The original executable sensitivity study at model/run_draft4_mediation_sensitivity.py is retained as an archival unit-level stress test. It generated five hypothetical causal patterns under high-fidelity, reference, and adverse measurement profiles at 40, 80, 160, and 320 independent units. At 320 units, the strong quantum-modulation scenario's mediated `Q -> N -> Y` route was detected in 99.7, 86.7, and 18.0 percent of simulations across those profiles. The weak route reached 40.7, 4.0, and 0.0 percent. In the Orch-OR-residual-candidate scenario, direct `Q -> Y | X,N,T` detection reached 95.3, 91.7, and 77.0 percent, while mediated detection reached 92.7, 65.0, and 7.0 percent. Classical-only and SAN-network-dominant scenarios kept direct quantum detection between 1.3 and 2.0 percent and mediated detection between 0.0 and 0.3 percent at the same sample size. Those results exposed measurement reliability as a design-critical variable but did not model nesting or missingness.

The final-preprint upgrade adds model/run_draft7_nested_mediation_sensitivity.py. It simulates 24 randomized repeated trials nested within each participant, participant heterogeneity, trial-order nuisance, separate measurement reliability for the quantum candidate, classical microtubule state, and neural mediator, and observed-variable-dependent missingness. The fixed seed is 20260728; each design cell contains 120 simulations. The null scenario is included to monitor false-positive behavior. The two prespecified quantum-route tests retain the family-wise `z=2.2414` threshold.

The variables are operational placeholders, not interchangeable labels. `Q_MT` is a prespecified microtubule quantum-candidate observable that must pass its own physical null and target-engagement assay. `X_MT` is a prespecified classical microtubule state or perturbation-engagement observable. `N` is a downstream neural mediator measured after `Q_MT` and `X_MT` but before the behavioral or report endpoint `Y`. A generic oscillation, energy-migration signal, coherence-loss curve, or anesthetic effect cannot be inserted into `Q_MT` merely because it is unusual.

At 80 participants under the reference profile, the simulation retained on average 84.5-85.0 percent of trials. The strong quantum-mediated scenario detected the indirect route in 100 percent of cells but a direct quantum residual in only 10.8 percent; the Orch-OR-residual scenario detected both the indirect and direct routes in 100 percent; the classical-only and SAN-network-dominant scenarios detected no indirect route and kept direct-quantum detection at 3.3 and 1.7 percent, respectively. The null scenario produced 0 percent indirect, 2.5 percent direct-quantum, and 2.5 percent network detections. Under the adverse profile, average retention fell to roughly 65 percent and the indirect route in the residual scenario fell to 79.2 percent, making the intended degradation visible.

These are synthetic detection frequencies, not empirical power, biological data, or evidence for either theory. Coefficients are hypothetical; Gaussian linear relations simplify biology; the cluster-robust within-participant model does not fit every random slope; the indirect-effect interval omits cross-equation covariance; and simulated missingness does not cover missing-not-at-random processes. A real registered protocol must replace all nuisance ranges with blinded preparation-specific estimates and rerun the frozen code before sample-size selection.

Detection of a residual term would still not establish objective reduction or constitutive consciousness. It would identify a candidate pathway requiring physical-witness validation, temporal precedence, classical-confound audits, replication, and content-level discrimination.

8.9 Registered-report analysis contract

The confirmatory protocol must identify one primary quantum route and one primary network route before unblinding. Exclusions are limited to prespecified acquisition failure, failed target engagement, verified artifact thresholds, or missing outcome rules; exclusions cannot depend on which theory a result favors. Multiplicity is controlled across the two primary quantum tests at family-wise alpha 0.05, while secondary frequency bands, lags, regions, and content classes are labeled exploratory and reported with false-discovery control and unthresholded effect intervals.

Robustness analyses include complete-case and inverse-probability-weighted estimates under the declared missing-at-random model; participant-level grouped resampling; alternative admissible lag windows; predeclared classical covariate sets; negative-control preparations and outcomes; and sensitivity to measurement reliability. The untouched biological-unit-grouped test set is opened once. Any theory-favoring interpretation fails if it depends on post-outcome windows, unregistered feature selection, one nuisance specification, leakage between units, or a physical measure that does not pass its own witness null.

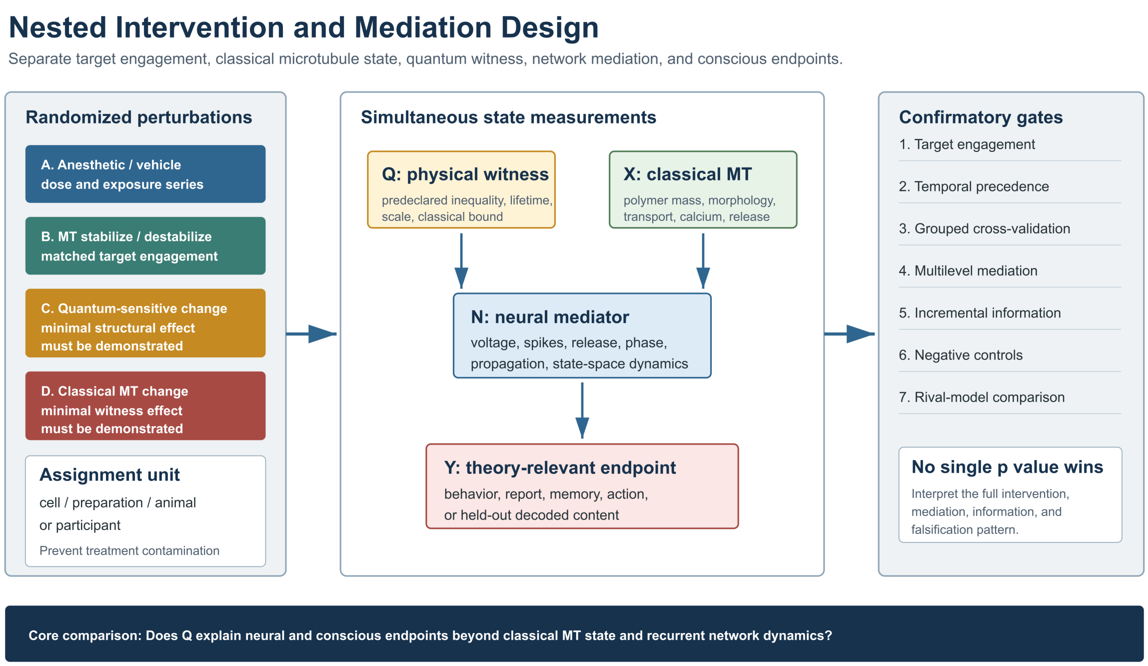

Figure 2. The experiment separates randomized perturbations, simultaneous state measurements, and confirmatory gates. The assignment unit must prevent contamination, and all validation splits remain grouped by biological unit. # 9. Relation to GNWT and other network theories

Criticism of one-frequency, point-neuron, or purely algorithmic abstractions is useful when it identifies a missing biological variable. It is not accurate to say that GNWT has no experimental evidence. A 2025 preregistered adversarial collaboration measured 256 participants with fMRI, MEG, and intracranial EEG and found a mixture of challenges and supportive findings for preregistered predictions [8]. Earlier and clinical work has also tested workspace-related ignition and long-range connectivity [9,10].

The stronger SAN comparison is therefore not "biology versus no evidence." It is whether SAN's dendritic, laminar, oscillatory, field, and sensorimotor variables explain findings that GNWT treats more abstractly and whether they make superior novel predictions. The same standard should be applied to Orch OR: biological richness is valuable only when it improves prediction and causal explanation.

The relevant classical comparator is also richer than a continuously active point-neuron workspace. Miller, Lundqvist, and Bastos synthesize working memory as sparse gamma bursts coordinated by alpha / beta control rather than uninterrupted persistent firing [29]. Sherfey and colleagues model how resonance, inhibition, and excitability can gate propagation from working-memory circuits toward action selection [30]. Those results do not establish a distributed conscious observer, PWD, neural rendering, or NAPOT. They provide concrete circuit-level nuisance and mediation variables that both Orch OR and SAN experiments must beat: burst rate, laminar phase, traveling-wave direction, inhibitory state, and propagation to action.

10. Discussion

The evidence supports a narrower and more productive conclusion than either categorical dismissal or declaration of proof. Microtubules are plausible contributors to consciousness-relevant neural function. Specific experiments report anesthetic-sensitive transport and collective optical phenomena. Animal perturbation studies indicate that microtubule state can modify anesthetic behavior. These observations justify sustained investigation.

They do not yet demonstrate objective reduction in vivo, show that reduction controls neural firing, map a reduction state to a conscious content, or establish that a reduction is the experience itself. Room-temperature quantum engineering in a synthetic gel strengthens the plausibility of warm organized quantum devices but does not transfer consciousness to the material by analogy.

SAN offers a different explanatory strategy. It locates the candidate observer in the distributed activity of the system rather than in a separate inner viewer or a single microscopic event. It attempts to connect perception to action through recurrent multiscale dynamics. Yet it must still demonstrate that its proposed tonic / phasic organization and phase-wave differentials are necessary, content-specific, and explanatorily superior.

The multisurface source recovery sharpens this conditional relationship without deciding it. Micah's recovered developmental position allows microtubules to act as classical intracellular routers or, conditionally, as quantum-sensitive modulators within a larger SAN loop. That hybrid is not a fifth validated theory. It inherits both burdens: an intracellular state must be measured and shown to alter neural dynamics, and the resulting network change must still explain content and action better than classical rivals. Conversely, a successful distributed network account would not show that microtubules are irrelevant; it would constrain their role to upstream support or modulation unless a constitutive residual survives.

The theories can therefore be compared without forcing premature exclusivity. A classical microtubule contribution may support SAN. A quantum microtubule contribution may modulate SAN. Orch OR becomes distinctive only when an objective-reduction-specific state is shown to be necessary for conscious moments or content beyond downstream neural organization.

11. Conclusion

The central scientific question is not whether microtubules matter or whether quantum effects can occur in warm matter. It is where the causal and constitutive work of consciousness is done. Answering that question requires a complete chain from physical state to neural transduction, distributed computation, behavior, and differentiable experience. Orch OR and SAN should be judged by the same mechanism-completion standard.

The historical comparison is atom-specific rather than all-or-nothing. Penrose and Hameroff retain priority on objective reduction, original Orch OR, and the early intracellular microtubule-control chain. Micah Blumberg retains the earlier public record on the two later target-relative composites established here: distributed spatial / dendritic rendering linked to action before Hameroff 2022, and increased event rate linked to subjective slowing before Hameroff 2023. Neither finding requires full NAPOT identity, and neither licenses a claim about copying or intent. Preserving both directions of priority produces a stronger testable paper than collapsing either program into one author-level verdict.

Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research presented in this paper. The underlying research program was not written or developed by AI. This body of work is Micah Blumberg's original research, developed through a continuous program spanning 2011-2026. Its central conceptual foundations were developed and recorded before ChatGPT's public release on November 30, 2022.

In later manuscript development, AI tools were used only in bounded supporting roles, including assistance with mathematical and computational formulation, code, grammar and spelling, and checks against medical, biological, and physics literature. These tools did not supply the original theories or authorship. Micah Blumberg selected the claims and sources, directed the analysis, approved the final language, and remains responsible for the paper's arguments, interpretations, and errors.

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Final-preprint status and remaining external review

All work within the manuscript-production lane is complete for this preprint version: the protected Draft 6 remains unchanged; the nested nuisance-and-missingness simulation runs deterministically; operational variables are frozen as assay roles rather than assumed instruments; exclusion, multiplicity, leakage, temporal-order, and robustness rules are explicit; primary chronology claims are bound to public Micah sources and final target papers; and the older-transcript risk is governed by original-audio verification.

Independent review by an Orch OR specialist, a SAN-critical systems neuroscientist, and a quantum-optics specialist remains desirable and may motivate a later revision. It is not represented as completed and is not treated as a blocker to posting this clearly labeled preprint. The paper reports no new biological data, pilot estimate, achieved empirical power, objective-reduction witness, access evidence, copying finding, or validation of either consciousness theory.